Software Test Plan (STP) - Online Quiz System

Project: Online Quiz System
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1. Introduction

Purpose: This document is the test plan for the Online Quiz System v1.0. It describes what will be tested, how, when and by whom, and lists the test cases used to verify the requirements in the SRS v1.1.

Scope: Testing covers the quiz flow (start, questions, answers, scoring, result, restart), answer validation, administrator question management (add, modify, delete), administrator login and access control, and the non-functional requirements. Internet, web and GUI features are not part of the system.

References: Online Quiz System SRS v1.1; Software Architecture & Design Specification v1.0; IEEE 829 Standard for Software Test Documentation.

Definitions: MCQ (Multiple Choice Question), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), TC (Test Case), Admin (Administrator), UI (User Interface).

2. Test Items

- Console UI - menus, questions, messages and results
- Quiz Controller / Quiz Logic - quiz flow, answer checking and scoring
- Admin Module - administrator login and question management (add, modify, delete)
- Question data file (questions.txt) and administrator credential file

3. Features to be Tested

Features mapped to SRS requirement IDs (all 26 requirements are tested):

- Quiz Operation: OQ-F-001 to OQ-F-011, OQ-F-016
- Question Management: OQ-F-012, OQ-F-013, OQ-F-014
- User and Access Control: OQ-F-015
- OQ-NF-001: Response time within 1 second
- OQ-NF-002: Score preserved throughout the quiz
- OQ-NF-003: Clear console messages
- OQ-NF-004: Admin data not exposed to normal users
- OQ-NF-005: Runs on a standard machine with a C/C++ compiler
- Security: OQ-SR-001 to OQ-SR-005 (section 5.1)

4. Features Not to be Tested

- Web, GUI and network features - not part of the system (SRS Constraints and Communications).

- The C/C++ compiler, standard libraries and operating system - third-party, assumed correct.
- Multi-user or simultaneous access - the system is a single-user console application.
- Hardware devices beyond a standard keyboard and display (SRS Hardware interfaces).

5. Test Approach / Strategy

Levels:

- Unit tests - each function (answer check, score update, input validation, login check, file read/write) is tested on its own.
- Integration tests - Console UI, Quiz Controller / Quiz Logic, Admin Module and the question file tested together.
- System tests - complete quiz and admin sessions run end-to-end on the console, by hand and with input files (./quiz < input.txt).
- Acceptance tests - demo to the instructor covering quiz flow, answer validation, scoring, question management, invalid input handling and administrator access (SRS section 6).

Types:

- Functional testing of every OQ-F requirement.
- Negative and boundary testing - invalid options (E, 5, symbols, empty, two letters), wrong or empty credentials, incomplete question data.
- Regression testing - all test cases re-run after every bug fix.
- Performance, reliability, usability and portability testing for the NFRs.
- Security testing (section 5.1).

Entry Criteria: Program compiles without errors; questions.txt with test questions and a test admin account are ready; test cases are reviewed.

Exit Criteria: All 26 test cases executed; all High-priority and security test cases passed; no open critical defect; RTM updated with results.

5.1 Security Validation

Each security requirement from SRS section 5.1 is checked. All security test cases must pass before release.

- OQ-SR-001 Login before question management - try add, modify and delete without logging in; a login prompt must appear first (TC-22).
- OQ-SR-002 Reject wrong credentials - wrong password, wrong username and empty fields must not give access (TC-23).
- OQ-SR-003 Validate admin input - empty question text, missing options and an answer outside A-D must be rejected and the file left unchanged (TC-24).
- OQ-SR-004 No direct admin menu - hidden or out-of-range menu numbers and words like "admin" must not open the admin menu (TC-25).
- OQ-SR-005 Hide credentials - the password must be masked while typing and never shown in any output (TC-26).
- Related checks: unauthenticated access to question management blocked (OQ-F-015, TC-15); correct answers and admin data not visible from the user menu (OQ-NF-004, TC-20).

- Code review: check that all input is read safely (no gets() or unbounded scanf) and that the password is never printed.

6. Test Environment

Hardware: Standard desktop or laptop PC with keyboard and display; no special hardware.

Software: Windows 10/11 with MinGW (GCC) and Linux (Ubuntu) with GCC; the Online Quiz System v1.0 build.

Tools: Console/terminal, input files for scripted runs, a stopwatch or timestamped test build for response time, spreadsheet or GitHub Issues for defect tracking.

Test Data: questions.txt with 5 questions and a known answer key; answer script answers_3of5.txt; a test admin account (test-only credentials); lists of invalid options and invalid question data.

7. Test Schedule

Milestones (planned; adjust to the course calendar):
- Test case design: 28-09-2026
- Environment setup: 03-Oct-2026
- Test execution start: 05-Oct-2026
- Test execution end: 18-Oct-2026
- Acceptance demo: 20-Oct-2026 to 23-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (section 13.2)
- Test Scripts (input files)
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	<Name>	Prepare plan, coordinate execution, update RTM
Test Engineer	<Name>	Design and execute test cases, log defects
Developer	<Name>	Support defect fixes and triage
Product Owner	<Name>	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in a stable build	Test modules separately as they are finished; request early builds
Console behaves differently on Windows and Linux	Run TC-21 early on both systems
Question file corrupted during admin tests	Keep a backup copy of questions.txt and restore it before each run
Limited time before the deadline	Run High-priority and security test cases first

11. Assumptions & Dependencies

- A C/C++ compiler is available on the test machines.
- The question file and test admin account are prepared before execution.
- Requirement IDs in the SRS v1.1 do not change during testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The program does not compile or crashes at start-up.
- A defect blocks more than 30% of the test cases (e.g. questions cannot be loaded).

Resume testing if:

- Blocking defects are fixed.
- The program builds and starts correctly.

13. Test Case Management & Traceability

The RTM maps every SRS requirement to its test case, using the same test case IDs as the SRS RTM. The Result column is updated after each test run.

13.1 Requirements Traceability Matrix

Req ID	Requirement	Module	Test case	Result
OQ-F-001	Start quiz	Quiz Controller / Quiz Logic	TC-01	Not run
OQ-F-002	One question at a time	Quiz Controller / Quiz Logic	TC-02	Not run
OQ-F-003	Four options A-D	Quiz Controller / Quiz Logic	TC-03	Not run
OQ-F-004	Accept answer	Quiz Controller / Quiz Logic	TC-04	Not run
OQ-F-005	Reject invalid option	Quiz Controller / Quiz Logic	TC-05	Not run
OQ-F-006	Check answer	Quiz Controller / Quiz Logic	TC-06	Not run
OQ-F-007	Score +1 if correct	Quiz Controller / Quiz Logic	TC-07	Not run
OQ-F-008	Score same if wrong	Quiz Controller / Quiz Logic	TC-08	Not run
OQ-F-009	Next question	Quiz Controller / Quiz Logic	TC-09	Not run
OQ-F-010	Final score	Quiz Controller / Quiz Logic	TC-10	Not run
OQ-F-011	Show final score	Quiz Controller / Quiz Logic	TC-11	Not run
OQ-F-012	Admin add question	Admin Module	TC-12	Not run
OQ-F-013	Admin modify question	Admin Module	TC-13	Not run
OQ-F-014	Admin delete question	Admin Module	TC-14	Not run
OQ-F-015	Block unauthenticated users	Admin Module	TC-15	Not run
OQ-F-016	Restart quiz	Quiz Controller / Quiz Logic	TC-16	Not run
OQ-NF-001	Response <= 1 s	Console UI / Core Logic	TC-17	Not run
OQ-NF-002	Score preserved	Console UI / Core Logic	TC-18	Not run
OQ-NF-003	Clear messages	Console UI / Core Logic	TC-19	Not run
OQ-NF-004	Admin data not exposed	Console UI / Core Logic	TC-20	Not run
OQ-NF-005	Portability	Console UI / Core Logic	TC-21	Not run
OQ-SR-001	Login before question management	Admin Module	TC-22	Not run
OQ-SR-002	Reject wrong credentials	Admin Module	TC-23	Not run
OQ-SR-003	Validate admin input	Admin Module	TC-24	Not run
OQ-SR-004	No direct admin menu	Admin Module	TC-25	Not run
OQ-SR-005	Hide credentials	Admin Module	TC-26	Not run

Coverage: 16/16 functional, 5/5 non-functional and 5/5 security requirements are covered (100%).

13.2 Test Cases

26 test cases: 16 functional, 5 non-functional and 5 security.

Test case	Req.	Preconditions and steps	Expected result
TC-01 Start a new quiz Functional, High	OQ-F-001	**Pre:** Program at main menu; questions.txt has 5 known questions. 1. Select Start Quiz from the main menu.	A new quiz starts, score is 0 and Question 1 is displayed.
TC-02 One question shown at a time Functional, High	OQ-F-002	**Pre:** Quiz started. 1. Observe the screen after the quiz starts. 2. Answer Question 1.	Only Question 1 is shown at first; Question 2 appears only after Question 1 is answered.
TC-03 Four options shown Functional, High	OQ-F-003	**Pre:** Quiz started. 1. Check each of the 5 questions.	Every question shows exactly four options labelled A, B, C and D.
TC-04 Valid answer accepted Functional, High	OQ-F-004	**Pre:** Quiz at Question 1. 1. Enter B.	B is accepted as the answer and the question is processed.
TC-05 Invalid option rejected Negative / Boundary, Medium	OQ-F-005	**Pre:** Quiz at Question 1. 1. Enter each of: E, 5, @, AB, empty input. 2. Then enter A.	Each invalid input shows an error message asking for A, B, C or D; Question 1 stays active and the score is unchanged; A is then accepted.
TC-06 Answer compared with key Functional, High	OQ-F-006	**Pre:** Quiz started; correct answer to Q1 is B and to Q2 is C. 1. Answer Q1 with B. 2. Answer Q2 with A.	Q1 is marked correct and Q2 is marked incorrect.
TC-07 Score increases on correct answer Functional, High	OQ-F-007	**Pre:** Quiz at Q1, score 0; correct answer is B. 1. Enter B.	Score becomes 1 (increases by exactly one).
TC-08 Score unchanged on wrong answer Functional, High	OQ-F-008	**Pre:** Quiz at Q2, score 1; correct answer is C. 1. Enter A.	Score stays 1.
TC-09 Moves to next question Functional, High	OQ-F-009	**Pre:** Quiz at Q1. 1. Enter an invalid option (E). 2. Enter a valid option (A).	After E the quiz stays on Q1; after A, Question 2 is displayed.
TC-10 Final score calculated Functional, High	OQ-F-010	**Pre:** Answer script answers_3of5.txt (3 correct, 2 wrong). 1. Complete the quiz using the answer script.	Final score equals the number of correct answers: 3.
TC-11 Final score displayed Functional, High	OQ-F-011	**Pre:** Quiz completed with 3 of 5 correct. 1. Observe the result screen.	The result screen clearly shows the final score, e.g. "Your score: 3 / 5".
TC-12 Admin adds a question Functional, Medium	OQ-F-012	**Pre:** Admin logged in with test credentials; questions.txt has 5 questions. 1. Choose Add Question. 2. Enter a valid question, four options and correct answer B. 3. Log out and start a quiz.	Question is saved to questions.txt (6 questions) and appears in the quiz.

Test case	Req.	Preconditions and steps	Expected result
TC-13 Admin modifies a question Functional, Medium	OQ-F-013	**Pre:** Admin logged in. 1. Choose Modify Question and select Q2. 2. Change the question text and correct answer. 3. Start a quiz.	The updated text and answer are saved and used in the quiz.
TC-14 Admin deletes a question Functional, Medium	OQ-F-014	**Pre:** Admin logged in; 6 questions stored. 1. Choose Delete Question and select Q6. 2. Confirm with Y. 3. Start a quiz.	Q6 is removed from questions.txt and no longer appears; 5 questions remain.
TC-15 Unauthenticated user blocked Negative / Security, High	OQ-F-015	**Pre:** Program at main menu; no admin logged in. 1. Try to open Add, Modify and Delete Question without logging in.	Each attempt shows "Access denied. Admin login required."; no question data changes.
TC-16 Restart after result Functional, Medium	OQ-F-016	**Pre:** Quiz completed with score 3/5. 1. Choose Restart on the result screen.	A new quiz starts from Question 1 with score reset to 0.
TC-17 Response time Performance, Medium	OQ-NF-001	**Pre:** Standard PC; 20 recorded inputs (menu choices and answers). 1. Enter each input and measure the time until the next screen appears (stopwatch or timestamps in a test build).	At least 95% of inputs (19 of 20) are processed within 1 second.
TC-18 Score preserved across runs Reliability, High	OQ-NF-002	**Pre:** Answer script answers_3of5.txt. 1. Run the quiz 3 times with the same answers.	The same score (3) is shown every time; score never resets or changes mid-quiz.
TC-19 Clear messages Usability, Medium	OQ-NF-003	**Pre:** 3 students who have not used the program. 1. Ask each to complete a quiz without any extra help. 2. Note any confusion.	Each user completes the quiz without extra instructions; all prompts say what input is expected.
TC-20 Admin data hidden from user menu Security, High	OQ-NF-004	**Pre:** Program at main menu as a normal user. 1. Go through every option in the user menu and the quiz. 2. Look for correct answers or question-management options.	No correct answers are shown before answering and no question-management function is reachable from the user menu.
TC-21 Runs on standard machines Portability, Medium	OQ-NF-005	**Pre:** Windows PC with MinGW (GCC) and Linux PC with GCC. 1. Compile the source on each machine. 2. Run one complete quiz.	Program compiles without errors and runs correctly on both, with no special hardware.
TC-22 Login required for question management Security, High	OQ-SR-001	**Pre:** Program at main menu; admin not logged in. 1. Select Admin. 2. Try Add, Modify and Delete.	A login prompt appears first; add, modify and delete are only available after successful login.

Test case	Req.	Preconditions and steps	Expected result
TC-23 Wrong credentials rejected Security / Negative, High	OQ-SR-002	**Pre:** Test admin account exists. 1. Log in with a wrong password. 2. Log in with a wrong username. 3. Log in with empty fields. 4. Log in with the correct credentials.	The first three attempts show "Invalid credentials" and give no access; only the correct credentials open the admin menu.
TC-24 Admin input validated Security / Negative, Medium	OQ-SR-003	**Pre:** Admin logged in. 1. Add a question with empty text. 2. Add a question with only 3 options. 3. Add a question with correct answer E.	Each is rejected with an error message; questions.txt is unchanged.
TC-25 No direct access to admin menu Security / Negative, High	OQ-SR-004	**Pre:** Normal user at main menu or in a quiz. 1. Enter hidden or out-of-range menu numbers (e.g. 0, 9, 99) and words such as "admin". 2. Enter the admin menu number during a quiz answer prompt.	Input is rejected as invalid; the admin menu never opens without login.
TC-26 Credentials not displayed Security, Medium	OQ-SR-005	**Pre:** Test admin account exists. 1. Type the password at the login prompt. 2. Log in with wrong and then correct credentials. 3. Check all screen output.	The password is masked (* or hidden) while typing and never appears in any message or output.

14. Test Metrics & Reporting

Metrics collected:
- % test cases executed
- % passed/failed
- Number of defects by severity
- Defect aging (days open)
- Requirement coverage

Reports:
- Execution status after each test run
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		